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SQIsign Midterm

INTRODUCTION

As the capabilities of quantum computing continue to evolve, modern encryption and authentication systems become increasingly less reliable. The systems being used currently rely on mathematical problems that are computationally infeasible to solve with classical computers. However, with the introduction of quantum algorithms, such as Shor's algorithm which is used to efficiently calculate the prime factors of large numbers, those once infeasible tasks can be solved exponentially faster. Algorithms like RSA or Diffie-Hellman would become obsolete in the future where large-scale quantum computers exist everywhere. Any information being kept secure with the previously mentioned algorithms could easily be decrypted using a quantum algorithm. This threat has led to the growing field of post-quantum cryptography (PQC), with a goal to create cryptographic schemes that can withstand attacks from both classical and quantum computing models.

One of the main goals of post-quantum computing is the development of quantum-safe digital signature schemes. Digital signatures are essential for verifying the authenticity, integrity, and non-repudiation of digital data. Digital signatures create the foundation of secure communication, financial systems, blockchain technology, and software verification. These classical signatures could easily be compromised by quantum attacks, leaving all digital data vulnerable, and trust in digital communications would be undermined. To prevent this, researchers and organizations all over the world are working to develop and standardize secure, efficient alternatives that can protect data from quantum attacks.

In response to the need for quantum-proof systems, the National Institute of Standards and Technology (NIST) initiated a global standardization project to evaluate and select quantum-resistant algorithms for future public use. This process involves multiple rounds of assessment to identify schemes that provide a balance between security, performance, and practicality. Several different mathematical approaches were explored regarding digital signatures, including lattice-based, hash-based, and isogeny-based constructions. This paper will focus on one of the isogeny-based systems called SQIsign (Short Quaternion and Isogeny Signature).

SQIsign was originally developed by a group of five cryptographers: Luca De Feo, David Kohel, Antonin Leroux, Christophe Petit, and Benjamin Wesolowski. In the latest NIST submission, there were over two dozen authors listed. SQIsign is a post-quantum digital signature scheme that gets its security from two computationally hard problems: (1) finding isogenies between supersingular elliptic curves, and (2) determining their endomorphism rings. It is believed that these problems will remain difficult, even for quantum computers, making SQIsign a strong candidate for long-term cryptographic security.

A central mathematical concept in SQIsign is the Deuring correspondence, which establishes a connection between elliptic curves and quaternion algebras. This correspondence allows operations performed on elliptic curves to be translated into equivalent operations within quaternion algebraic structures. It creates an efficient mathematical framework that supports compact data representation and secure computation. By using this correspondence, SQIsign is able to achieve one of the smallest key and signature sizes among all post-quantum signature proposals.

Compactness is one of SQIsign's defining advantages. Many post-quantum algorithms produce very large keys and signatures that can reach several kilobytes in size, creating storage and transmission challenges. SQIsign, in contrast, has a public key size of approximately 65 bytes and a signature size of around 148 bytes. These small parameters make it especially suitable for environments where efficiency is crucial, including Internet of Things (IoT) devices, embedded systems, smart cards, and mobile platforms. Smaller signatures also reduce bandwidth consumption and verification time, which benefits applications such as blockchain networks, where digital signatures are generated and validated frequently.

The internal structure of SQIsign follows the same general model used in most digital signature systems but incorporates advanced mathematical mechanisms to provide post-quantum security. The algorithm operates through three core phases: key generation, signing, and verification. During key generation, a secret isogeny is selected, and the associated elliptic curve becomes the public key. In the signing phase, the signer constructs a proof demonstrating knowledge of the secret isogeny without revealing it. This is done using a non-interactive zero-knowledge proof based on the Fiat-Shamir transform, which ensures the proof remains secure while eliminating the need for interaction between parties. The verifier then checks that the proof correctly corresponds to the public key and message, confirming both authenticity and validity.

In addition to efficiency, SQIsign exhibits strong resilience against known attacks. Earlier isogeny-based systems, such as SIKE (Supersingular Isogeny Key Encapsulation), were eventually compromised through advances in cryptanalysis that exploited weaknesses in their mathematical structures. SQIsign, however, relies on a different set of assumptions involving quaternion algebras, which provide greater resistance to the types of attacks that broke SIKE. So far, no known classical or quantum algorithms have been able to efficiently solve the problems upon which SQIsign's security depends.

Beyond its direct cryptographic value, SQIsign contributes to the overall diversity and strength of the post-quantum cryptographic landscape. Many algorithms currently under evaluation by NIST rely on lattice-based assumptions, which, while well studied, introduce the risk of a single point of failure if those assumptions are ever weakened. SQIsign, built on an entirely different mathematical foundation, provides an independent line of defense. This diversity ensures that global security is not overly dependent on one cryptographic paradigm.

SQIsign represents the successful application of advanced mathematical theory to practical cybersecurity challenges. By combining supersingular isogeny theory, quaternion algebra, and non-interactive proof systems, it achieves a rare balance of security, compactness, and efficiency. Its design not only addresses the immediate need for quantum-safe digital signatures but also demonstrates how deep mathematical insights can lead to practical, long-lasting cryptographic solutions. As the world moves toward the quantum era, SQIsign stands out as one of the most promising candidates for securing digital communication and authentication in future generations of technology.

BACKGROUND & PRELIMINARIES

To understand SQIsign, one must first be familiar with the core mathematical constructs it is built upon. SQIsign's security relies on complex mathematical constructs that are believed to be resistant to quantum attacks. This section provides a detailed overview of these foundations, focusing on the concepts that are

critical to understanding the scheme's internal operations, particularly the Deuring correspondence and the algorithms that leverage it.

1. Elliptic Curves

An elliptic curve is a smooth, projective algebraic curve defined by an equation of the form $y^2 = x^3 + ax + b$ over a field K , where the discriminant is non-zero. In cryptography, we typically work over finite fields F_p or F_{p^2} . The set of points on an elliptic curve, including a special "point at infinity," forms an abelian group (where the group operation is commutative, so the order of elements doesn't change the result). This group structure is the basis for Elliptic Curve Cryptography (ECC), where the security relies on the hardness of the Elliptic Curve Discrete Logarithm Problem (ECDLP).

2. Supersingular Elliptic Curves

A special class of elliptic curves, known as supersingular curves, is crucial for isogeny-based cryptography. Over a finite field of characteristic p , a supersingular elliptic curve can be defined as one whose endomorphism ring is a maximal order in a quaternion algebra— in other words, the complexity of supersingular curves makes it difficult for even quantum computers to figure them out, making them perfect for cryptography. A more accessible characterization is that the number of points on a supersingular curve E over F_p is congruent to 1 modulo p . These curves are not vulnerable to certain attacks that affect ordinary curves and, most importantly, provide the rich mathematical structure that SQIsign exploits.

3. Isogenies

An isogeny $\phi : E_1 \rightarrow E_2$ is a non-constant rational map between two elliptic curves that is also a group homomorphism. If an isogeny exists between two curves, they are called isogenous. The degree of an isogeny is its degree as an algebraic map. Isogenies can be composed, and for every isogeny, there exists a unique dual isogeny. Crucially, for a fixed prime ℓ , the set of supersingular elliptic curves over F_{p^2} and the ℓ -isogenies (isogenies of degree ℓ) between them form a connected, Ramanujan graph. This graph is known as the *supersingular isogeny graph*. Navigating this graph is computationally difficult, which forms the basis of security.

4. The Hard Problems

The security of SQIsign rests on the computational hardness of the following problems, which are believed to be intractable even for large-scale quantum computers:

- **Supersingular Isogeny (SSI) Problem:** Given two supersingular elliptic curves E_1 and E_2 defined over a finite field, find an isogeny $\phi : E_1 \rightarrow E_2$ of degree d between them. In the

context of SQIsign, the secret key is this isogeny path, and forging a signature is equivalent to finding a new, specific isogeny path.

- Endomorphism Ring Problem: The endomorphism ring of an elliptic curve, denoted $\text{End}(E)$, is the ring of all isogenies from E to itself. For a random supersingular elliptic curve E , computing its entire endomorphism ring is a problem considered exponentially hard for both classical and quantum computers. SQIsign's security is closely tied to the difficulty of computing endomorphism rings, as knowledge of $\text{End}(E)$ allows for efficient computation of isogenies from E .

5. The Deuring Correspondence

A pivotal concept that enables SQIsign's efficiency is the *Deuring correspondence*. This is a one-to-one correspondence between:

- The set of supersingular elliptic curves over F_p^2 , and
- The set of maximal orders in the quaternion algebra $B_{p,\infty}$ ramified at p and ∞ .

This bijection allows cryptographic operations defined in the world of elliptic curves and isogenies to be translated into equivalent, and often more efficient, operations in the world of quaternion algebras. SQIsign leverages this correspondence to perform critical computations, which is a key reason for its compact signature and key sizes compared to other post-quantum schemes.

6. From Interactive to Non-Interactive Proofs

SQIsign is built around a zero-knowledge proof of knowledge, where the signer proves they know the secret isogeny (or equivalently, the endomorphism ring of their public key curve) without revealing it. Initially, this is an interactive protocol between a "prover" (signer) and a "verifier." To convert this into a non-interactive digital signature that can be verified by anyone at any time, SQIsign employs the **Fiat-Shamir transform**. This transformation replaces the verifier's random challenge with a cryptographic hash of the message combined with the prover's initial commitment. The resulting proof is the digital signature.

With these concepts in mind—supersingular elliptic curves, isogenies, the hardness of finding them, the Deuring correspondence, and the Fiat-Shamir transform—we can now proceed to a detailed description of the SQIsign scheme itself.

DESCRIPTION OF THE SCHEME

SQIsign is an isogeny-based post-quantum signature protocol built upon the arithmetic of supersingular elliptic curves and the hardness of computing their endomorphism rings. It was developed as part of ongoing research into quantum-safe cryptography, addressing the need for secure digital signatures that can withstand attacks from quantum computers. Traditional cryptographic systems such as RSA and ECC

(Elliptic Curve Cryptography) depend on problems like integer factorization or the discrete logarithm, both of which can be efficiently solved by Shor's algorithm on a large-scale quantum computer. To ensure long-term digital security, new cryptographic systems must rely on mathematical problems that remain hard even for quantum computation. SQIsign achieves this by using isogeny-based cryptography, one of the most mathematically compact and elegant families of post-quantum schemes.

At its core, SQIsign's idea is to represent cryptographic keys as elliptic curves and to encode the private key as a secret isogeny path - a structure that connects two supersingular elliptic curves. Each node in the isogeny graph represents an elliptic curve, and each edge corresponds to an isogeny, which is a special type of algebraic map that preserves the group structure of elliptic curves. The public key is one endpoint of this path (a curve that can be publicly known), while the private key is the secret knowledge of how to travel between these curves via the isogeny.

The scheme constructs a zero-knowledge proof of knowledge that shows the signer knows the secret path (or equivalently, the endomorphism ring of the elliptic curve) without revealing any information about it. This interactive proof is then converted into a non-interactive signature protocol through the Fiat-Shamir transformation, replacing the verifier's challenge with a hash of the message and the commitment. As a result, the final digital signature is both secure and compact.

What makes SQIsign particularly notable is its signature size. With signatures as small as 128-148 bytes and public keys around 64 bytes, it is currently one of the most compact post-quantum signature schemes proposed. This efficiency makes SQIsign well suited for environments where memory and bandwidth are limited, such as IoT devices, embedded processors, and secure smartcards.

In summary, SQIsign uses the mathematical hardness of finding isogenies between supersingular elliptic curves to create a quantum-resistant digital signature mechanism that remains practical for real-world deployment.

The security of SQIsign depends on several key mathematical hardness assumptions that have been rigorously studied in the field of isogeny-based cryptography.

1. **Supersingular Isogeny Problem (SSI):**

Given two supersingular elliptic curves E_1 and E_2 defined over a finite field F_p , it is computationally infeasible to find an isogeny $\phi: E_1 \rightarrow E_2$ connecting them. In the context of SQIsign, the private key corresponds to such a secret isogeny path, and the security of the scheme depends on the difficulty of recovering that path.

2. **Endomorphism Ring Problem:**

For a supersingular elliptic curve, E , the set of all isogenies from E to itself forms a mathematical object known as the endomorphism ring, denoted $End(E)$. Determining $End(E)$ for a random supersingular curve is an extremely hard computational problem, and no efficient (classical or quantum) algorithm is known to solve it. In SQIsign, the signer's private key is related to knowledge of this endomorphism ring, which enables efficient isogeny computation.

3. **KLPT Framework (Kohel–Lauter–Petit–Tignol Algorithm):**

SQIsign leverages the KLPT algorithm, which provides a way to move between quaternion

ideals and elliptic curve isogenies. This algorithmic bridge enables practical implementation of isogeny computations while preserving theoretical hardness.

4. **Hardness Against Quantum Attacks:**

While quantum algorithms can break RSA and ECC, no known quantum algorithm can efficiently solve the isogeny or endomorphism ring problems. This makes SQIsign **post-quantum secure**, meaning it should remain safe even in the presence of powerful quantum computers.

Together, these assumptions ensure that forging signatures, deriving private keys, or impersonating a valid signer is computationally infeasible within realistic time and resource constraints.

In practice, SQIsign follows three main stages: **key generation**, **signing**, and **verification**.

1. **Key Generation:**

- a. Select a base supersingular elliptic curve E_0 .
- b. Compute a secret isogeny ϕ that connects E_0 to a new curve E_A .
- c. The public key is the curve E_A ; the private key is the isogeny ϕ .

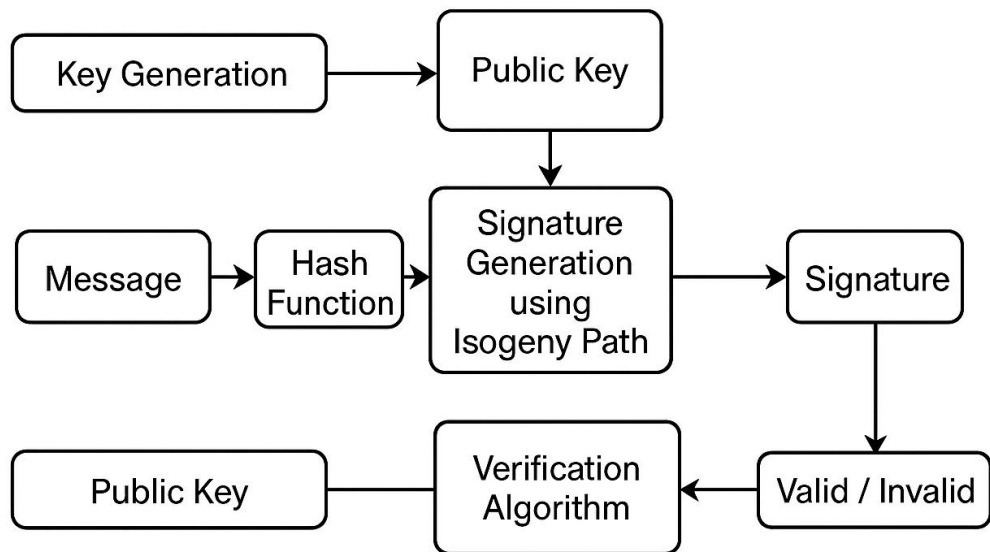
2. **Signing Process:**

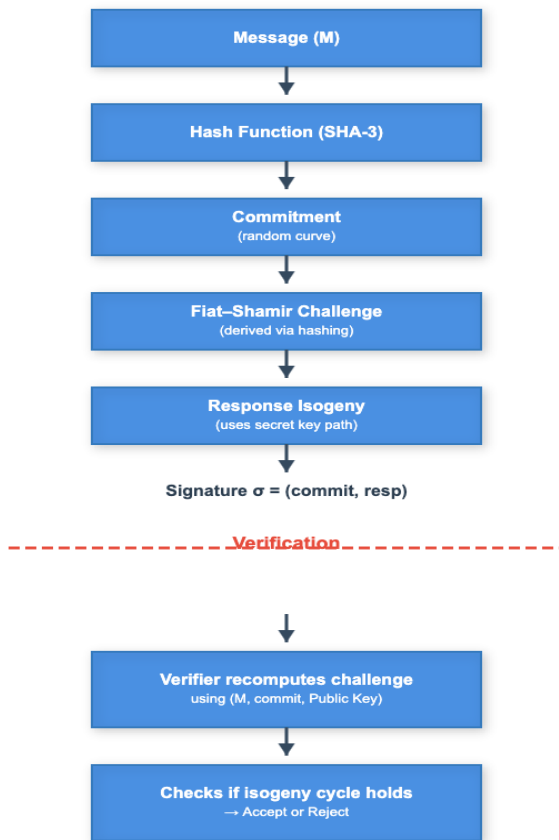
- a. The signer hashes the message M using a cryptographic hash function (e.g., SHA-3).
- b. They create a commitment by generating a temporary random isogeny and computing its corresponding elliptic curve.
- c. A deterministic “challenge” is derived by hashing together the commitment, public key, and message.
- d. Using the secret key, the signer computes a response isogeny that satisfies the challenge.
- e. The resulting triple (commitment, challenge, response) forms the signature.

3. **Verification:**

- a. The verifier uses the public key and message to recompute the expected challenge.
- b. They verify that the isogeny relationships in the signature form a consistent cycle in the isogeny graph.
- c. If all checks pass, the signature is accepted as valid.

Below is a conceptual flow diagram describing the SQIsign process.





ALGORITHMS

SQIsign is an isogeny-based post-quantum digital signature scheme that relies on the hardness of the *supersingular isogeny problem*. It uses links called isogenies, which are mappings between a special type of elliptic curves called *supersingular elliptic curves* defined over the finite field $\mathbb{F}_p^2$. SQIsign uses advanced mathematical techniques to produce public keys and compact signatures while still maintaining strong post-quantum security standards.

Supersingular Isogeny Problem

The backbone of SQIsign is its reliance on the *supersingular isogeny problem*. This problem uses *supersingular elliptic curves* instead of ordinary elliptic curves, which adds an extra layer of mathematical complexity and security. A *supersingular elliptic curve* is a special kind of elliptic curve with additional structure (internal connections or “bridges”), called *endomorphism rings*. If we think of isogenies as bridges connecting one elliptic curve to another, then we can think of *endomorphism rings* as

looping bridges that start and end on the same elliptic curve. These loops create additional structures on elliptic curves, making it harder to reverse the relationships between curves. The supersingular isogeny problem goes as follows:

- Given two supersingular elliptic curves.
- Find the bridge (the specific isogeny) that connects them.

Even though the base (starting point) elliptic curve and the signer's public key are both known, it is still extremely difficult to determine the exact isogeny (bridge linking them together) that maps elliptic curves. This hardness is the foundation of SQIsign's security and what allows it to be a post-quantum cryptographic system.

Key Generation

The signer starts from a fixed base elliptic curve E_0 over the finite field $\mathbb{F}_p^2$, E_0 is publicly known, and will be the non-changing starting point for everyone using this crypto system. The signer then chooses a secret isogeny $\phi: E_0 \rightarrow E_a$ of degree $\ell_a^{e_a}$ (ℓ_a is a small prime number, e_a is an integer exponent). This secret isogeny ϕ is the signer's private key.

From this connection, the public elliptic curve $E_a = \phi(E_0)$ is defined, and two reference points P^B and Q^B on E_0 (both points on the starting elliptic curve) are mapped to E_a as $\phi(P^B)$ and $\phi(Q^B)$. The curve E_a is also public (like E_0), but each person signing a message has a unique E_a . Each signer's private isogeny allows them to generate a unique public curve E_a . The public key consists of $(E_a, \phi(P^B), \phi(Q^B))$. In short:

- Private key = the secret isogeny ϕ
- Public key = $(E_a, \phi(P^B), \phi(Q^B))$

Signing

To sign a message m , the signer hashes the message to produce $H(m)$. This hash creates a unique elliptic curve for the message being signed, call it E_m . Using the secret isogeny ϕ (the signer's private key), the signer then builds a link-like structure of small-degree isogenies that link E_a to the newly created curve E_m :

$$E_a \rightarrow E_1' \rightarrow E_2' \rightarrow E_3' \rightarrow E_m$$

(The E_i' curves are intermediate elliptic curves along the path from E_a to E_m , and the arrows are isogenies of small degree).

This chain of isogenies and supersingular elliptic curves forms the signature. One of the key features that makes SQIsign so powerful is that the signature and public key remain small—only 204 bytes—while maintaining strong security properties.

Verification

The verifier receives:

- The message m
- The signature (the isogeny path $E_a \rightarrow E_m$)
- The signer's public key $(E_a, \phi(P^B), \phi(Q^B))$, where P^B and Q^B are points)

Using these, the verifier performs the following steps to confirm that the message and signature are authentic:

1. Verification:

- Re-hash the message to get $H(m)$: The verifier re-hashes the message to make sure it's the same message that the signer originally signed.
- Use the public curve E_a and the public points $\phi(P^B)$, $\phi(Q^B)$ to rebuild the curve that should match the signer's E_m ; we'll call the verifier's version E'_m .
- Removes E_m that's inside the signature.
- Check if the curve E'_m (built by the verifier) is the same as E_m (the curve that came with the message).
- If E'_m equals E_m , the signature is valid.

Verification (Pseudocode)

1. input: message m , signature $sign$, public key $(E_a, \phi(P^B), \phi(Q^B))$
2. $Hm = \text{Hash}(m)$
3. $E'_m = \text{RebuildCurveFromPublicData}(E_a, \phi(P^B), \phi(Q^B), H(m))$
4. $E_m = \text{ExtractCurveFromSignature}(sign)$
5. if $E'_m == E_m$:
 - return 'ACCEPT (valid signature)'else:
 - return 'REJECT (invalid signature)'

Check

1. Re-hash message: $Hm = \text{Hash}(m)$
2. Recompute expected curve: $E'_m = \text{RebuildCurveFromPublicData}(E_a, \phi(P^B), \phi(Q^B), Hm)$
3. Compare with the signer's curve: if $E'_m == E_m$, the signature is valid

CRYPTANALYSIS

IMPLEMENTATION – Runtime and Tradeoffs

1. Runtime Performance and Complexity Analysis

The practical deployment of SQIsign requires careful consideration of its computational performance, which is intrinsically tied to its complex mathematical operations. Runtime is dominated by isogeny and

endomorphism ring computations, which are significantly more expensive than operations in classical or lattice-based schemes.

- **Key Generation:** This is the most computationally intensive phase, typically requiring several seconds on a standard CPU (e.g., 2-10 seconds for NIST security levels). The process involves:
 - Generating a random secret isogeny from a public starting curve E_0 .
 - Computing the corresponding public curve E_A and the public torsion basis images $\phi(PB)$, $\phi(QB)$.
 - Critically, it requires the computation of the endomorphism ring of E_A using the *KLPT algorithm* (Kohel–Lauter–Petit–Tignol). This algorithm, which navigates the quaternion algebra via the Deuring correspondence to find an equivalent ideal and translate it back to an isogeny, involves solving a norm equation and performing intensive ideal operations. Its complexity is polynomial but with large constants, making it the primary bottleneck.
- **Signing:** The signing operation is also computationally heavy, often comparable to or slightly faster than key generation (1-8 seconds). It involves:
 - Hashing the message to derive a "challenge" isogeny kernel.
 - Computing a response isogeny path that connects the public key curve to a curve derived from this challenge. This again requires the application of the KLPT algorithm (or a variant thereof) to construct an isogeny satisfying specific degree and embedding conditions. The need for precise, structured isogeny computation, rather than random walks, dictates this high cost.
- **Verification:** Verification is notably faster than signing or key generation, typically in the range of 10-100 milliseconds. It involves:
 - Reconstructing the challenge from the message and signature commitment.
 - Composing several provided isogenies (the signature) and checking that they form a cycle with the public key, ending at the correct curve. This process primarily requires elliptic curve scalar multiplications and isogeny evaluations, which are relatively efficient and parallelizable.

Summary: SQIsign exhibits an asymmetric runtime profile: **slow key generation, slow signing, but fast verification**. This profile is inverted compared to many lattice-based schemes, which often feature fast key generation and signing but slower verification.

2. Critical Trade-offs

The design of SQIsign embodies several fundamental trade-offs that distinguish it from other post-quantum candidates.

- **Compactness vs. Computational Cost:** This is the central trade-off. SQIsign achieves unparalleled compactness (~64-byte public keys, ~150-byte signatures) by leveraging the rich, high-dimensional structure of supersingular isogeny graphs and the Deuring correspondence. However, harnessing this structure for efficient, *directed* computation (as required for signing) is inherently complex. The algorithms (KLPT) that enable this compact representation are algebraically intensive. In contrast, lattice-based schemes like Dilithium have larger sizes (e.g., 2-

4 kB combined) but simpler, linear-algebraic operations that are faster and easier to implement in constant-time.

- **Algorithmic Agility vs. Implementation Footprint:** SQIsign's security relies on a unique mathematical foundation (hard problems on supersingular isogenies and endomorphism rings). This provides valuable cryptographic diversity. However, the implementation footprint is large and complex. It requires:
 - A full stack of arithmetic over quadratic extension fields (F_{p^2}).
 - Sophisticated elliptic curve point arithmetic and isogeny evaluation (vélu-style formulas).
 - The entire machinery of the KLPT algorithm in quaternion algebras.This complexity increases the risk of subtle implementation bugs, makes side-channel resistance more challenging to achieve, and results in a larger codebase compared to hash-based or some code-based schemes.
- **Determinism vs. Performance Optimization:** The signing algorithm in SQIsign is largely deterministic, driven by the hash of the message. This eliminates risks associated with poor randomness during signing (a critical failure point in ECDSA/RSA). However, it also limits opportunities for precomputation or caching to speed up the signing operation, as each signature requires a fresh execution of the core response algorithm.
- **Verifier Efficiency vs. Prover Burden:** The fast verification property is a major advantage for protocols where one party signs once and many parties verify (e.g., software updates, certificate validation). This trade-off deliberately burdens the signer (who is often a server with more resources) to benefit the verifiers (which may be numerous or constrained). This makes SQIsign particularly appealing for applications like blockchain, where a transaction is signed once but validated by every node.
- **Maturity and Cryptanalysis Resilience vs. Optimization Potential:** SQIsign is a relatively newer scheme compared to lattice-based finalists. Its core mathematical components have undergone significant revision and optimization since its initial proposal (e.g., the "SQIsignHD" improvements). While this demonstrates active and fruitful cryptanalysis, it also means that implementation best practices and optimization techniques are still evolving. Future research may substantially improve runtime (as seen with the move from "SQIsign" to "SQIsignHD"), but current implementations represent a point on an improving curve, not a final plateau.

Implementing SQIsign requires accepting high computational cost for signing and key generation in exchange for minimal bandwidth and storage requirements. Its fast verification is a standout feature. The primary challenges are:

1. Mastering and correctly implementing the deep mathematical stack, particularly the KLPT algorithm.
2. Managing the performance bottleneck of signing, which may limit its use in high-throughput signing scenarios without dedicated hardware acceleration.
3. Ensuring constant-time execution across all layers (field arithmetic, isogeny computation, ideal arithmetic) to protect against timing attacks.

The ideal use cases are in environments where signature size is paramount, verification speed is critical, and the signer's computational burden is acceptable—such as in embedded systems with intermittent

signing needs, certificate authorities, or blockchain protocols. For applications requiring millions of signatures per second, lattice-based schemes currently offer a more practical performance profile, albeit with larger data sizes.

REFERENCES

Aardal, M. A., Adj, G., Aranha, D. F., Basso, A., Canales Martínez, I. A., Chávez-Saab, J., Corte-Real Santos, M., Dartois, P., De Feo, L., Duparc, M., Eriksen, J. K., Fouotsa, T. B., Gazzoni Filho, D. L., Hess, B., Kohel, D., Leroux, A., Longa, P., Maino, L., Meyer, M., Nakagawa, K., Onuki, H., Panny, L., Patranabis, S., Petit, C., Pope, G., Reijnders, K., Robert, D., Rodríguez-Henríquez, F., Schaeffler, S., & Wesolowski, B. (2025). SQIsign [NIST Post-Quantum Cryptography submission]. National Institute of Standards and Technology. <https://SQIsign.org>

De Feo, Luca, David Kohel, Anthony Leroux, Christophe Petit, and Benjamin Wesolowski. “SQIsign: Compact Post-Quantum Signatures from Quaternion Isogenies.” *Advances in Cryptology – CRYPTO 2023*, Springer, 2023, pp. 64–95.

“Post-Quantum Cryptography Project.” *National Institute of Standards and Technology (NIST)*, <https://csrc.nist.gov/Projects/post-quantum-cryptography>.

PQC Round 4 Submissions Overview. National Institute of Standards and Technology, 2023.

“Supersingular Isogeny Key Exchange.” *Wikipedia*: Wikimedia Foundation, 2024, https://en.wikipedia.org/wiki/Supersingular_isogeny_key_exchange.

“A Survey on Various Stream Authentication Techniques.” *International Journal of Engineering Research & Technology (IJERT)*, <https://www.ijert.org/a-survey-on-various-stream-authentication-techniques>.

SQIsign Team. (2023). *SQIsign Specification (Version 2023-06-01)*. SQIsign Official Website. Retrieved from <https://sqisign.org/spec/sqisign-20230601.pdf>